

Music Genre Classification Using Machine Learning

This project presents a complete machine learning pipeline for automatic music genre classification using engineered audio features. The workflow includes preprocessing, train-validation-test splitting, visualization, model development, hyperparameter tuning, evaluation, feature importance analysis, and final locked test assessment.

Project Structure

EDA and Preprocessing Notebook for cleaning and exploratory analysis. **Modeling and Evaluation Notebook** for training, tuning, and testing. **features_3_sec_cleaned.csv** as the cleaned dataset used for modeling. **Scaled and tree-based train/test datasets** for different model types. **Saved outputs** including plots, CSV reports, and trained models.

Implemented Models

Three machine learning models were implemented and compared: **Logistic Regression** as a fast and interpretable baseline. **Random Forest** to capture nonlinear feature interactions. **Neural Network (MLPClassifier)** for deeper nonlinear learning. Hyperparameter tuning was performed using validation data while preserving the integrity of the locked test set.

Evaluation Pipeline

The evaluation process included: Accuracy, Precision, Recall, Macro F1-score, and Weighted F1-score. Confusion matrices and normalized confusion matrices. Per-class F1-score analysis. Feature importance interpretation. Validation versus test performance comparison.

Academic Integrity

The project follows proper academic machine learning methodology: The dataset was cleaned before training. The test set remained completely unseen during preprocessing and tuning. Validation data was used for model selection decisions. The final test evaluation was performed only once. All artifacts were saved to ensure reproducibility.

Generated Outputs

Artifact	Description
final_test_results.csv	Final test metrics
final_classification_report.csv	Detailed evaluation report
feature_importance.csv	Feature ranking analysis
baseline_model_comparison.png	Baseline comparison plots
tuned_model_comparison.png	Tuned model comparison plots
final_selected_model.joblib	Serialized final model

This project demonstrates a complete supervised machine learning workflow for multiclass music genre classification. The final pipeline is suitable for academic reporting, GitHub publication, and machine learning portfolio presentation.